The chain

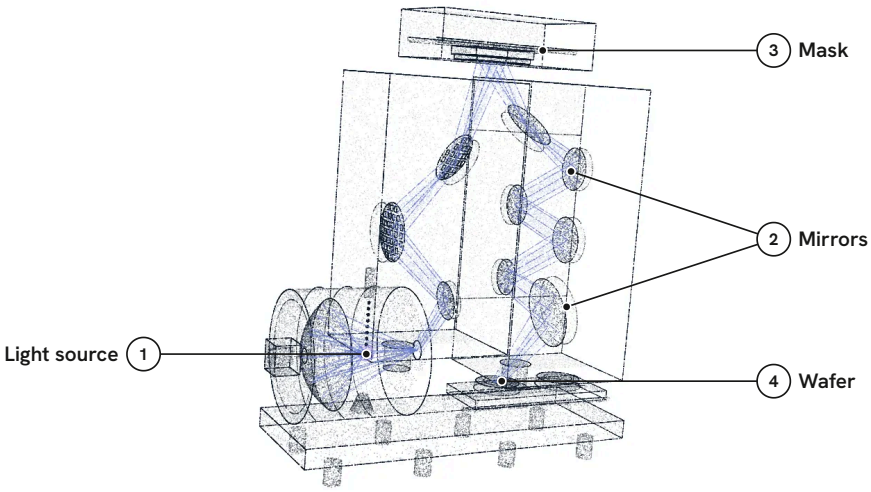
01 Chip Design, EDA and IP
02 Silicon and Wafers
03 Chemicals, Gases and Photoresist
04 Lithography
05 Photomasks and Pellicles
06 Deposition and Etch
07 Metrology and Inspection
08 Foundries and Fabs
09 Memory and HBM
10 Advanced Packaging
11 Substrates and PCBs
12 Test and Assembly
13 Systems and Networking
14 Data Centers and Power

15 The Economics of AI Chips
16 Who Controls the Chain

Stage 4 of 14 · Extreme concentration · Very hard to substitute

Lithography

ASML, a Dutch firm, is the only maker of extreme ultraviolet scanners, the machines that print the finest layers of a chip. It sold 48 of them in 2025.



Specimen 04 · Lithography

An extreme ultraviolet scanner cut open along its light path. The machine is the size of a bus and weighs about 180 metric tons. The blue-violet thread is the 13.5 nm light. The mask and the wafer are drawn about twice their true size.

Parts, as numbered on the drawing

1. Light source: ASML San Diego (US), laser by Trumpf (DE)
2. Mirrors: Zeiss SMT (DE)
3. Mask: see stage 05, Photomasks and Pellicles
4. Wafer: see stage 02, Silicon and Wafers

In plain terms

Lithography prints the pattern of a circuit onto the wafer. A machine holds a stencil of one layer of the circuit up to a light source and projects the image, four times smaller, onto the silicon. The shorter the wavelength of the light, the finer the lines a machine can print. Even so, a machine can print lines narrower than its light's wavelength. Immersion scanners, for example, shine the light through a layer of water between lens and wafer, which lets them print smaller features with the same light[1]. ASML rates one of its immersion scanners, the

1 Lenses & mirrors - Lithography principles. ASML, accessed 9 September 2026. asml.com

NXT:2150i, at features of 38 nanometers or less, printed with argon fluoride laser light at 193 nanometers[2]. Air soaks up extreme ultraviolet light, the shortest in use, so machines that print with it work in a vacuum. Only ASML, a Dutch company, makes them. Export controls are government rules on who a company may sell to. Every advanced chip passes through ASML's machines, so export controls on ASML affect all advanced chipmaking.

In short

Beyond extreme ultraviolet (EUV), ASML holds above 80 percent of the market for deep ultraviolet (DUV) lithography, which uses light of longer wavelength[3]. In July 2026 ASML put its capacity for 2026 at around 65 standard EUV scanners, which caps how many fabs anywhere can add that year[4]. ASML says the production capacity of Zeiss, its sole supplier of optics, limits how many lithography systems it can build[5].

Export controls set up by the United States with allies including the Netherlands block EUV scanners from Chinese chip foundries[6]. In December 2025 Reuters reported a Chinese prototype that had made EUV light but had printed no chip[7]. In July 2026 Reuters reported, citing one unnamed source, that a Shanghai firm with two state shareholders has started producing immersion DUV tools, which shine their light through water[8].

ASML sold 327 lithography systems in 2025[9]. Tools shipped to China made up 33 percent of its net system sales that year, more than any other region[10].

Chokepoint card Lithography

Concentration

Extreme

Three or fewer firms could supply this stage today, or one firm holds more than half of it.

Substitutability

Very hard

ASML is the only maker of EUV scanners, and Zeiss is its sole supplier of their mirrors[5].

Price or market size

About $200 million

for a standard EUV scanner, an estimated $350 million to $400 million for the newer High-NA version and roughly $60 million for a tool in the DUV segment, according to Reuters in July 2026[3]

2 TWINSCAN NXT:2150i. ASML, accessed 18 October 2026. asml.com
3 Explainer: The $400 million ASML 'printers' key for the AI chip boom. Reuters, 28 July 2026. reuters.com
4 Press Release Financial Results Q2 2026. ASML, 15 July 2026. ourbrand.asml.com
5 ASML Holding, 2025 annual report, filed on Form 6-K. U.S. Securities and Exchange Commission, 25 February 2026. sec.gov
6 Breakthroughs or Boasts? Assessing Recent Chinese Lithography Advancements. Center for Strategic and International Studies, 24 September 2025. csis.org
7 How China built its 'Manhattan Project' to rival the West in AI chips. Reuters, 17 December 2025. reuters.com
8 China starts production of home-grown immersion DUV chipmaking tools, source... Reuters, 28 July 2026. reuters.com
9 Financial performance: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 5. ourbrand.asml.com
10 ASML 2025 fourth-quarter and full-year results (investor presentation). ASML, 28 January 2026, p. 9. ourbrand.asml.com

Who leads	NL	**ASML**	327 lithography systems sold in 2025, including all 48 EUV (extreme ultraviolet) systems [9]
	JP	**Canon**	234 chip lithography systems sold in 2025 [11]
	JP	**Nikon**	22 new chip lithography systems sold in the year to March 2026 [12]
Where it is made	NL	**Netherlands**	ASML, the Dutch maker of every EUV scanner
	DE	**Germany**	Zeiss SMT optics, Oberkochen, and Trumpf carbon dioxide drive lasers, Ditzingen
	US	**United States**	ASML light-source research and manufacturing, San Diego
	JP	**Japan**	Nikon and Canon lithography scanners and steppers
Why substitution is slow	A newcomer would first have to make mirrors a meter wide, polished smooth to within tens of picometers, a picometer being a trillionth of a meter [1]. It would also need a light source that hits tin droplets with a laser 50,000 times a second [13].		

Where China stands

In July 2026 Reuters reported, citing one unnamed source, that a Shanghai firm with two state shareholders had started producing immersion DUV tools [8]. The Center for Technology & Statecraft found in September 2026 that Chinese firms had begun small-batch production of such scanners, but none had yet been shown fit for producing advanced chips [14]. Its central estimate in 2026 was that China would have a commercially viable immersion scanner for 7 nm chips in the mid-2030s, with the early 2030s plausible, and EUV not before the second half of the 2030s [14]. Meanwhile Chinese fabs had built a stockpile of more than 330 immersion DUV systems by the first quarter of 2026, most of them ASML's NXT:1980i [14].

Where the US stands

CSET found in 2021 that no American firm made lithography scanners [15]. In September 2026 the Center for Technology & Statecraft found that ASML held about 99 percent of the market for immersion DUV scanners and Japan's Nikon the rest [14]. But every immersion DUV scanner holds a light source from Cymer, a US company now part of ASML, or from Japan's Gigaphoton, so the two governments could stop sales by controlling those parts, the Center for Technology & Statecraft found in September 2026 [14]. A US rule published in October 2023 already covers immersion scanners with an alignment error between 1.5 nm and 2.4 nm bound for advanced chipmaking if they hold any American-made parts [16]. The US entity list also requires a license to supply the firms it names [17].

01

How lithography prints a chip's pattern

The stencil is called a photomask, and the machine a scanner (see Photomasks and pellicles).

The wafer is coated with photoresist, a film that changes wherever light touches it. The mask holds one layer of the circuit as a pattern of clear and dark areas, drawn at four times its size on the chip. The machine shines light through the mask, shrinks the image four times with a lens, and lands it on the film.

A wash then removes the film where the light hit, or, with some films, where it did not, leaving the pattern standing on the wafer [18]. The film then serves as a stencil for the next step, which either etches the pattern into the layer beneath or fires ions into the open areas to change how the silicon conducts [18]. Then the film is stripped and the next

1 Lenses & mirrors - Lithography principles. ASML, accessed 9 September 2026. asml.com

8 China starts production of home-grown immersion DUV chipmaking tools, source... Reuters, 28 July 2026. reuters.com

9 Financial performance: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 5. ourbrand.asml.com

11 Canon Inc. FY 2025 Results. Canon, 29 January 2026, p. 22. global.canon

12 Financial Results: The Year Ended March 31, 2026. Nikon, 8 May 2026, p. 8. nikon.com

13 All about light and lasers in lithography. ASML, accessed 9 September 2026. asml.com

14 DUV Immersion Lithography: The Foundation of the US AI Advantage. Center for Technology & Statecraft, September 2026. techstatecraft.org

15 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 4. cset.georgetown.edu

16 Export Controls on Semiconductor Manufacturing Items. Federal Register, 25 October 2023. federalregister.gov

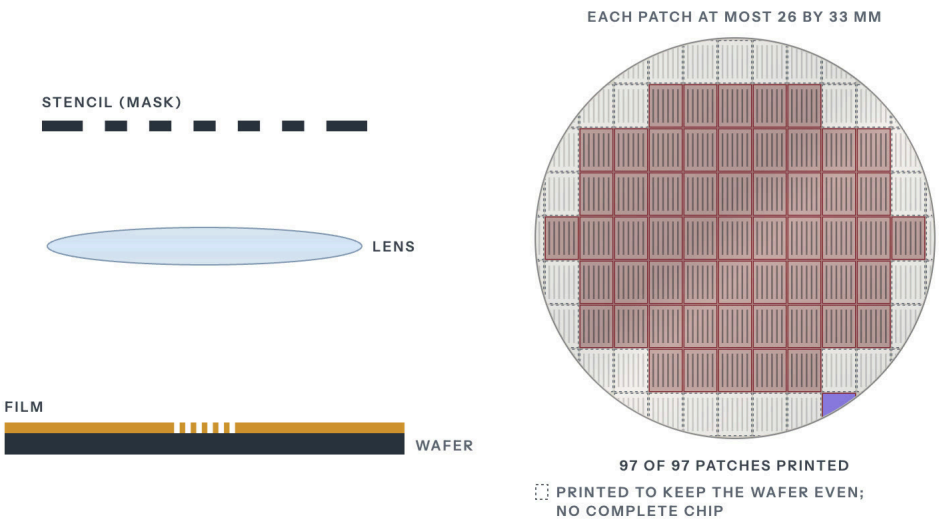
17 Expansion of End-User Controls To Cover Affiliates of Certain Listed Entities. Federal Register, 30 September 2025. federalregister.gov

18 6 crucial steps in semiconductor manufacturing. ASML, 4 October 2023. asml.com

mask goes in. Chips are made up of dozens of layers, ASML says[18]. Photronics, a US mask maker, says masks are typically sold in sets, with each layer's pattern on a different mask[19].

Figure 04.1 · Explainer, 19 seconds

How lithography prints a chip



1. **Coat.** The wafer is coated with a thin film that changes wherever light touches it.
2. **Stencil.** A stencil, called a mask, holds the pattern for one layer of the chip, drawn four times larger than life.
3. **Shrink.** Light shines through the stencil, and a lens shrinks the image four times and lands it on the film.
4. **Wash.** A wash removes the film wherever the light landed, and the pattern is left on the wafer.
5. **Repeat.** The machine steps across the wafer, printing one patch at a time, each about the size of a postage stamp.

Note: Simplified. The newest machines, which use extreme ultraviolet light, carry the image with mirrors in place of a lens.

The image from one mask covers a patch of up to 26 by 33 mm[20]. The wafer moves under the optics one patch at a time, close to a hundred patches per wafer. ASML's own wafers-per-hour rating for an EUV scanner assumes 96 of them[20]. Before the wafer is exposed, the machine measures alignment marks printed in earlier layers so the new layer lines up with them[20]. For its NXE:3400B EUV scanner, ASML rates the matched-machine overlay, a measure of that alignment, at 2 nm[20]. That model first shipped in 2017[21].

Finer lines need shorter light. Glass and air both absorb the shortest light in use, at 13.5 nm[22]. So an EUV scanner works in a vacuum and uses mirrors instead of lenses, and its masks are mirrors too[22]. Deep ultraviolet (DUV) machines use light with a

18 6 crucial steps in semiconductor manufacturing. ASML, 4 October 2023. asml.com

19 Photronics, annual report on Form 10-K for the fiscal year ended 31 October 2025. U.S. Securities and Exchange Commission, 17 December 2025. sec.gov

20 TWINSCAN NXE:3400B: EUV lithography systems. ASML, accessed 27 September 2026. asml.com

21 Stronger than expected demand drives ASML Q1 sales. U.S. Securities and Exchange Commission, 19 April 2017. sec.gov

22 ASML EUV lithography systems. ASML, accessed 9 September 2026. asml.com

wavelength more than fourteen times longer[13]. ASML calls its immersion DUV scanners the workhorses of the industry, used in high-volume manufacturing of the most advanced logic and memory chips[23].

When one exposure cannot draw a pattern finely enough, chipmakers use multipatterning. In the litho-etch kind, the pattern is split between two or more masks printed in turn[24]. Each added mask brings more process steps, more cost and a risk that the parts of the pattern misalign[25]. Self-aligned double and quadruple patterning (SADP and SAQP) print the lines with one exposure and then multiply them with deposition and etch steps[24]. In SAQP each printed line ends up as four[24]. Because one exposure places all those lines, they cannot misalign with each other[24]. A later exposure still prints blocks that break them where the circuit needs gaps[24]. ASML says one EUV exposure per layer cuts masks and process steps and lowers cost[26].

Figure 04.2 · Explainer, 26 seconds

How several passes turn one line into four



1. **Print.** A deep ultraviolet scanner shines light through a mask and prints a row of lines, called cores, in one exposure. The chip needs lines four times as dense, closer than one exposure can draw.
2. **Coat.** A thin film is deposited over everything, covering the top and both sides of every core.
3. **Etch back.** An etch straight down clears the film from the tops and the floor and leaves it on the sides of each core. These strips are the spacers, and the edges of the cores set where they stand.
4. **Two lines.** An etch removes the cores and leaves the spacers standing, two for each printed line. All of them trace back to one exposure, so they cannot misalign with each other.

13 All about light and lasers in lithography. ASML, accessed 9 September 2026. asml.com

23 See ASML's DUV lithography systems. ASML, accessed 9 September 2026. asml.com

24 EUV lithography: weighing the options. Imec, 6 June 2019. imec-int.com

25 Applied Materials' Innovative Pattern-Shaping Technology Reduces the Cost... Applied Materials, 28 February 2023. ir.appliedmaterials.com

26 Strategic report: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 16. ourbrand.asml.com

5. **Four lines.** The spacers now serve as cores. The same deposition and etch steps give each of them spacers of its own, so every printed line ends up as four, at a quarter of the spacing.
6. **Cut.** A later exposure prints blocks that break the lines where the circuit needs gaps. ASML says one EUV exposure per layer cuts masks and process steps and lowers cost.

Note: Simplified. Drawn in cross-section, as if the wafer were cut across the lines. The deep ultraviolet mask is drawn at the size of the pattern it prints.

02

DUV and EUV scanners compared

Deep ultraviolet

Deep ultraviolet (DUV) light comes from gas lasers called excimer lasers: 248 nanometers from krypton fluoride (KrF), 193 from argon fluoride (ArF), shone through air or water[13]. Cymer, an independently run ASML business in San Diego, makes these excimer lasers, which generate the light that scanners use to print patterns on wafers[27]. Gigaphoton, a Komatsu company in Japan, also makes excimer lasers for lithography tools[28].

Of the systems ASML sold in 2025, 131 were ArF immersion, 78 KrF and 16 ArF dry, all deep ultraviolet, and another 54 were older i-line tools[10]. The i-line tools use 365 nm light from a mercury lamp, and ASML says they print features as small as 220 nm[13].

Reuters gives one price for the whole DUV segment, with tools costing roughly $60 million each[3]. ASML's own figures split its DUV sales by tool type. In 2025 the 131 ArF immersion tools it sold made up 42 percent of its system sales, and the 78 KrF tools only 4 percent[10]. In DUV, ASML competes with Nikon and Canon, and it still holds above 80 percent of that market[3].

Chipmakers without EUV build their finest layers on immersion DUV scanners, in several passes per layer (figure 04.2)[26]. From 1 September 2023, Dutch export licensing covered the most advanced of these scanners, ASML's NXT:2000i and later models[29].

Extreme ultraviolet

The 13.5 nm light comes from molten tin droplets shot from a generator[13]. A laser hits each droplet twice, first with a low-intensity pulse that flattens it and then with a more powerful one that vaporizes it into plasma[13]. The process repeats 50,000 times a second[13]. Mirrors made by Zeiss, a German optics company, carry the light to the wafer[1]. The largest are a meter across and smooth to within tens of picometers, a picometer being a thousandth of a nanometer[1].

1 Lenses & mirrors - Lithography principles. ASML, accessed 9 September 2026. asml.com
3 Explainer: The $400 million ASML 'printers' key for the AI chip boom. Reuters, 28 July 2026. reuters.com
10 ASML 2025 fourth-quarter and full-year results (investor presentation). ASML, 28 January 2026, p. 9. ourbrand.asml.com
13 All about light and lasers in lithography. ASML, accessed 9 September 2026. asml.com
26 Strategic report: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 16. ourbrand.asml.com
27 Explore ASML San Diego. ASML, accessed 9 September 2026. asml.com
28 Gigaphoton, company profile. Gigaphoton, accessed 9 September 2026. gigaphoton.com
29 Statement regarding export control regulations Dutch government. ASML, 30 June 2023. asml.com

Brighter light means more wafers an hour, and ASML demonstrated the first 1,000-watt EUV source in April 2025[30]. Its TWINSCAN NXE:3800E ships at its full 220 wafers an hour, 37 percent more than the NXE:3600D[31].

High-NA EUV

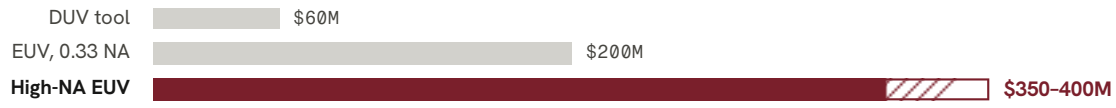
High-NA scanners raise the numerical aperture (NA), a measure of how well the optics collect and focus light, from 0.33 to 0.55[32]. That cuts the smallest printable feature from 13 nm to 8 nm[32]. Their optics shrink the mask image four times in one direction and eight in the other, halving the patch of wafer one exposure covers and doubling the exposures per wafer[32]. The second-generation TWINSCAN EXE:5200B runs at 175 wafers an hour, 60 percent above the EXE:5000[26].

Reuters puts a standard EUV tool at around $200 million and a High-NA machine at an estimated $350 to $400 million[3].

Figure 04.3

High-NA EUV scanners cost an estimated $350 million to $400 million in 2026

What one lithography system costs. The DUV bar is one figure for the whole segment



Source: Reuters explainer on ASML's lithography machines, first published 28 January 2026 and updated 28 July 2026 [3]

ASML sold four High-NA (EXE) systems in 2025, against two in 2024[33]. In its 2025 annual report, filed in February 2026, ASML said it expected High-NA to start supporting high-volume manufacturing in 2027[5].

Intel and TSMC have chosen differently on High-NA:

- **Intel Foundry** says it uses High-NA in volume production for select layers of some Intel 18A products[34]. It installed the first EXE:5200B[35]. By September 2026 it had run more than one million wafers on High-NA tools, a count that includes tool testing and research as well as volume production[34].
- **TSMC** waited, and in September 2026 said it will use High-NA in high-volume manufacturing from 2030[36].

On 8 September 2026 ASML and TSMC announced a move from 6-inch photomasks to 12-inch ones, to remove stitching constraints[36]. Stitching constraints are the design limits that come from printing a large chip in two halves. ASML and TSMC aim for a

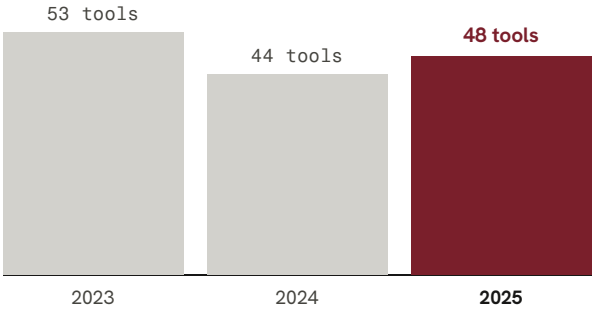
3 Explainer: The $400 million ASML 'printers' key for the AI chip boom. Reuters, 28 July 2026. reuters.com
5 ASML Holding, 2025 annual report, filed on Form 6-K. U.S. Securities and Exchange Commission, 25 February 2026. sec.gov
26 Strategic report: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 16. ourbrand.asml.com
30 Strategic report: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 30. ourbrand.asml.com
31 Strategic report: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 17. ourbrand.asml.com
32 5 things you should know about High NA in EUV. ASML, 25 January 2024. asml.com
33 Financial performance: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 7. ourbrand.asml.com
34 Intel Foundry and ASML Accelerate Industry Readiness for High-NA EUV. Intel, 7 September 2026. intel.com
35 High NA EUV reaches new readiness milestone with first high-volume Logic product. ASML, 15 July 2026. asml.com
36 ASML and TSMC Announce Initiative to Pioneer Industry Transition to... TSMC, 8 September 2026. pr.tsmc.com

pilot line for the new masks by 2031 and for High-NA systems that use them to enter production by 2033[36].

Figure 04.4

ASML sold 48 EUV systems in 2025, up from 44 in 2024

EUV systems ASML sold each year



Source: ASML 2025 annual report for 2024 and 2025, and ASML's Q4 2023 investor presentation for 2023 [9]

03

Who makes lithography machines

ASML's net sales were EUR 32.7 billion in 2025, at a 52.8 percent gross margin[9]. In October 2026 it expected 2026 net sales of EUR 44 to 45 billion, at a gross margin of around 56 percent[37]. Its 2025 annual report lists 5,100 suppliers[5]. Three of them and ASML's own San Diego site make the main parts of the EUV machine:

- **Zeiss SMT**, Oberkochen, Germany, makes the illumination system and the six-mirror projection optics[38].
- **Trumpf**, Ditzingen, Germany, makes the carbon dioxide drive laser that hits the tin droplets[39]. It amplifies a few watts to 40 kilowatts[39].
- **ASML San Diego**, which includes the light-source maker Cymer that ASML bought in 2013, designs the source where the laser turns tin into plasma, and builds the droplet generator[27].
- **VDL ETG** builds the frames that suspend and position the mirrors[40].

ASML says the production capacity of Zeiss, its sole supplier of optics, limits how many lithography systems it can build[5].

Lithography is the stage of the chain that European firms lead, with ASML the only maker of EUV scanners[26]. The bulk of ASML's suppliers are in the Netherlands and the

5 ASML Holding, 2025 annual report, filed on Form 6-K. U.S. Securities and Exchange Commission, 25 February 2026. sec.gov
9 Financial performance: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 5. ourbrand.asml.com
26 Strategic report: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 16. ourbrand.asml.com
27 Explore ASML San Diego. ASML, accessed 9 September 2026. asml.com
36 ASML and TSMC Announce Initiative to Pioneer Industry Transition to... TSMC, 8 September 2026. pr.tsmc.com
37 ASML reports €11.7 billion total net sales and €5.0 billion net income in Q3... U.S. Securities and Exchange Commission, 14 October 2026. sec.gov
38 EUV lithography and technology. Carl Zeiss, accessed 9 September 2026. zeiss.com
39 Good things come in ever-smaller packages. TRUMPF, accessed 9 September 2026. trumpf.com
40 VDL ETG builds complex frames for Zeiss. VDL Groep, 22 March 2019. web.archive.org

rest of Europe, the Middle East and Africa, the company says[41].

In Japan, Nikon and Canon also make lithography tools. Nikon sold 22 new chip lithography systems in the year to March 2026, in a market it sizes at 570 units for 2025[12]. Its Precision Equipment business made an operating loss in the year to March 2026[42]. Canon sold 234 chip lithography systems in 2025, 188 of them i-line and 46 KrF[11]. Its range runs from KrF scanners to i-line steppers, including some built for advanced packaging[43]. Canon also makes a nanoimprint system, the FPA-1200NZ2C, which stamps the pattern instead of projecting it and can print lines down to 14 nm wide[44]. Canon delivered one in 2024 to the Texas Institute for Electronics, to be used for research and prototypes[44].

In January 2021 CSET, a research center at Georgetown University, found that the United States lacked firms in a few key parts of the chip supply chain, above all lithography tools[15].

Figure 04.5

EUV made up 48 percent of ASML's 2025 system sales

ASML net system sales by technology, 2025



Source: ASML Q4 2025 investor presentation [10]

04

Lithography export controls and China's own scanners

Export controls set up by the United States with allies including the Netherlands block ASML from shipping EUV scanners to Chinese foundries[6]. In September 2025 CSIS judged that EUV remained "firmly out of reach" for China despite massive state investment[6]. For DUV scanners and other tools, the restrictions came in this order:

- **Dutch national licensing** of advanced DUV took effect on 1 September 2023[45].

6 Breakthroughs or Boasts? Assessing Recent Chinese Lithography Advancements. Center for Strategic and International Studies, 24 September 2025. csis.org

10 ASML 2025 fourth-quarter and full-year results (investor presentation). ASML, 28 January 2026, p. 9. ourbrand.asml.com

11 Canon Inc. FY 2025 Results. Canon, 29 January 2026, p. 22. global.canon

12 Financial Results: The Year Ended March 31, 2026. Nikon, 8 May 2026, p. 8. nikon.com

15 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 4. cset.georgetown.edu

41 Sourcing & Supply Chain. ASML, accessed 27 October 2026. asml.com

42 Financial Results: The Year Ended March 31, 2026. Nikon, 8 May 2026, p. 6. nikon.com

43 Semiconductor Lithography Equipment. Canon, accessed 18 October 2026. global.canon

44 Canon delivers FPA -1200NZ2C nanoimprint lithography system for semiconductor... Canon, 26 September 2024. global.canon

45 The Netherlands expands export control measure for advanced semiconductor... Government of the Netherlands, 6 September 2024. government.nl

- **A US rule** published on 25 October 2023 covered immersion scanners that print features of 45 nm or smaller with an alignment error between 1.5 nm and 2.4 nm[16]. If such a scanner was bound for advanced chipmaking, the rule put it under US export rules when it held any American content at all[16]. The rule did not apply to scanners first exported from a country that lists them on its own export control list[16].
- **A partial license revocation** by the Dutch government, disclosed by ASML on 1 January 2024, stopped shipments of NXT:2050i and NXT:2100i immersion scanners to a few Chinese customers[46].
- **The Netherlands widened its licensing** to more types of deep ultraviolet lithography equipment from 7 September 2024[45]. On 15 January 2025 it announced that from 1 April 2025 the license requirement would also cover a very limited number of further goods, among them specific measuring and inspection equipment[47].
- **The BIS Affiliates Rule**, issued by the US Bureau of Industry and Security, extended entity-list controls to affiliates owned 50 percent or more by listed entities from 29 September 2025[17]. BIS then suspended it until 9 November 2026[48]. Exporting goods covered by US export rules to a firm on the entity list needs a US license[17].
- **The EU** added certain Dutch controls to the EU Control List in November 2025[5]. ASML says this changed nothing for its immersion tools, because they already needed a Dutch license[5].

China's share of ASML's net system sales fell from 41 percent in 2024 to 33 percent in 2025[10].

On DUV, Reuters reported in July 2026, citing one unnamed source, that Shanghai Aishengna Electronic Technology Group has started producing home-grown immersion DUV tools[8]. The firm was registered in August 2023 with 7 billion yuan of capital and two state shareholders, Shanghai Electric Holding and a Shanghai International Trust subsidiary[8]. It has no website and has taken in teams from SMEE, China's older scanner maker, and the startup Yuliangsheng[8]. SMEE is strongest in i-line, the coarsest kind of lithography tool, and even there holds only four percent of the world market[6].

The Center for Technology & Statecraft, a research group, found in September 2026 that none of China's home-grown immersion scanners had yet been shown fit for producing advanced chips[14]. Its central estimate was a commercially viable Chinese immersion

5 ASML Holding, 2025 annual report, filed on Form 6-K. U.S. Securities and Exchange Commission, 25 February 2026. sec.gov

6 Breakthroughs or Boasts? Assessing Recent Chinese Lithography Advancements. Center for Strategic and International Studies, 24 September 2025. csis.org

8 China starts production of home-grown immersion DUV chipmaking tools, source... Reuters, 28 July 2026. reuters.com

10 ASML 2025 fourth-quarter and full-year results (investor presentation). ASML, 28 January 2026, p. 9. ourbrand.asml.com

14 DUV Immersion Lithography: The Foundation of the US AI Advantage. Center for Technology & Statecraft, September 2026. techstatecraft.org

16 Export Controls on Semiconductor Manufacturing Items. Federal Register, 25 October 2023. federalregister.gov

17 Expansion of End-User Controls To Cover Affiliates of Certain Listed Entities. Federal Register, 30 September 2025. federalregister.gov

45 The Netherlands expands export control measure for advanced semiconductor... Government of the Netherlands, 6 September 2024. government.nl

46 Statement regarding partial revocation export license. ASML, 1 January 2024. asml.com

47 Klever: export controls on advanced semiconductor manufacturing equipment to be... Government of the Netherlands, 15 January 2025. government.nl

48 One Year Suspension of Expansion of End-User Controls for Affiliates of Certain... Federal Register, 12 November 2025. federalregister.gov

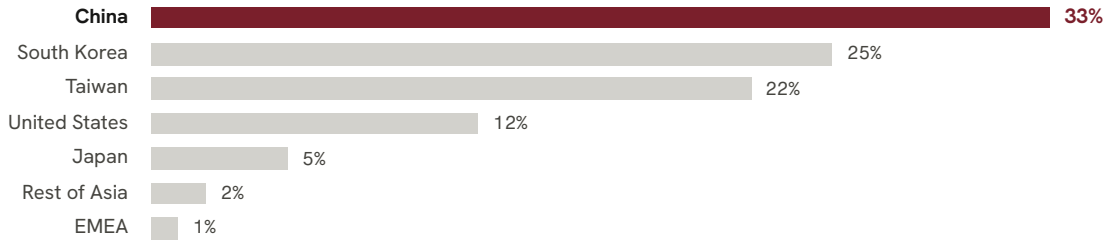
scanner for 7 nm chips in the mid-2030s, and EUV not before the second half of the 2030s[14]. Until then Chinese fabs rely on imported tools, and by the first quarter of 2026 they had built a stockpile of more than 330 immersion DUV systems, most of them ASML's NXT:1980i[14].

There is less evidence of Chinese progress on EUV. Reuters reported in December 2025 that a Shenzhen team of former ASML engineers had built a prototype that made extreme ultraviolet light but had printed no chip[7]. To print a chip, a machine also needs mirrors that carry that light to the wafer, and ASML's sole supplier of those mirrors is Zeiss[5].

Figure 04.6

China took 33 percent of ASML's 2025 system sales, the largest share

ASML net system sales by the region tools shipped to, 2025



Source: ASML Q4 2025 investor presentation [10]

Note: Net system sales leave out service and field option sales, which ASML counts in its total net sales. The Export Controls page measures China's share against total net sales.

05

Why ASML's EUV scanners are a chokepoint

No other company makes EUV scanners, so every fab that adds EUV capacity buys it from ASML[26]. The scanners already in fabs keep printing only as long as they are serviced. ASML has around 10,000 customer support employees who maintain and upgrade those machines around the clock[49]. Serviced this way, ASML's machines stay in use for a long time. In its 2025 annual report ASML said about 95 percent of the systems it had sold in the previous 30 years were still in use[5]. The same report warned that export controls may limit the service ASML can provide for machines already installed[50].

ASML has a chokepoint of its own. Zeiss SMT is its only supplier of lenses, mirrors and other critical optics[51]. ASML says that if Zeiss could not supply them for a long period, ASML would effectively be unable to run its business[51].

5 ASML Holding, 2025 annual report, filed on Form 6-K. U.S. Securities and Exchange Commission, 25 February 2026. sec.gov

7 How China built its 'Manhattan Project' to rival the West in AI chips. Reuters, 17 December 2025. reuters.com

10 ASML 2025 fourth-quarter and full-year results (investor presentation). ASML, 28 January 2026, p. 9. ourbrand.asml.com

14 DUV Immersion Lithography: The Foundation of the US AI Advantage. Center for Technology & Statecraft, September 2026. techstatecraft.org

26 Strategic report: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 16. ourbrand.asml.com

49 Strategic report: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 40. ourbrand.asml.com

50 Strategic report: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 65. ourbrand.asml.com

51 Strategic report: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 68. ourbrand.asml.com

ASML's capacity caps how many EUV scanners fabs anywhere can add each year. In July 2026 ASML put its 2026 capacity at around 65 standard (low-NA) EUV scanners and said it plans to add 30 percent for 2027[4].

Leading-edge AI accelerators are made with EUV, which prints their finest layers, the ones that set how densely transistors and wires pack[22]. TSMC expects the number of layers needing High-NA to rise, driven mainly by the more complex transistors that AI chips need[36].

Four questions to answer from memory

1. What does lithography do? `see How it works`
2. Why do chipmakers print the finest layers with extreme ultraviolet (EUV) light? `see How it works`
3. Who makes EUV machines? `see Controls and China's answer`
4. What limits how many EUV machines chip factories can add each year? `see The chokepoint`

Four things to remember

- Lithography prints each layer of a chip's circuit onto the wafer.
- Extreme-ultraviolet (EUV) light prints the finest layers in fewer steps than deep ultraviolet, but air and glass absorb it, so EUV machines work in a vacuum and use mirrors.
- ASML of the Netherlands is the only maker of EUV machines, and Zeiss alone supplies their mirrors.
- Export controls block EUV sales to Chinese chipmakers, and a Chinese prototype reported in December 2025 had printed no chip.

Sources (51)

1. A Lenses & mirrors - Lithography principles. ASML · Accessed 9 September 2026
   `www.asml.com/en/technology/lithography-principles/lenses-and-mirrors`
2. A TWINSCAN NXT:2150i. ASML · Accessed 18 October 2026
   `www.asml.com/en/products/duv-lithography-systems/twinscan-nxt-2150i`
3. B Explainer: The $400 million ASML 'printers' key for the AI chip boom. Reuters · 28 July 2026
   `www.reuters.com/world/asia-pacific/250-million-asml-printer-behind-nvidias-chips-2026-07-28/`
4. A Press Release Financial Results Q2 2026. ASML · 15 July 2026
   `ourbrand.asml.com/asset/c8dbf3fc-4c5e-4406-83f6-27694b138245/Press-Release-Financial-Results-Q2-2026.pdf`
5. A ASML Holding, 2025 annual report, filed on Form 6-K. U.S. Securities and Exchange Commission (filing by ASML Holding) · 25 February 2026
   `www.sec.gov/Archives/edgar/data/937966/000162828026011377/asml-2025xannualxreportx.htm`
6. A Breakthroughs or Boasts? Assessing Recent Chinese Lithography Advancements. Center for Strategic and International Studies · 24 September 2025
   `www.csis.org/blogs/strategic-technologies-blog/breakthroughs-or-boasts-assessing-recent-chinese-lithography`
7. B How China built its 'Manhattan Project' to rival the West in AI chips. Reuters · 17 December 2025
   `www.reuters.com/world/china/how-china-built-its-manhattan-project-rival-west-ai-chips-2025-12-17/`
8. B China starts production of home-grown immersion DUV chipmaking tools, source says. Reuters · 28 July 2026
   `www.reuters.com/world/china/china-starts-production-home-grown-immersion-duv-chipmaking-tools-source-2026-07-28/`

4 Press Release Financial Results Q2 2026. ASML, 15 July 2026. `ourbrand.asml.com`

22 ASML EUV lithography systems. ASML, accessed 9 September 2026. `asml.com`

36 ASML and TSMC Announce Initiative to Pioneer Industry Transition to... TSMC, 8 September 2026. `pr.tsmc.com`

9. A Financial performance: From ASML's Annual Report 2025. ASML · 24 February 2026 · p. 5
ourbrand.asml.com/m/419103cb23dfeaa4/original/asml-2025-annual-report-financial-performance-section.pdf#page=5
10. A ASML 2025 fourth-quarter and full-year results (investor presentation). ASML · 28 January 2026 · p. 9
ourbrand.asml.com/m/3136300aa4999bc1/original/2026_01_28_Presentation-Investor-Relations-Q4-2025.pdf#page=9
11. A Canon Inc. FY 2025 Results. Canon · 29 January 2026 · p. 22
global.canon/en/ir/conference/pdf/conf2025e.pdf#page=22
12. A Financial Results: The Year Ended March 31, 2026. Nikon · 8 May 2026 · p. 8
www.nikon.com/content/dam/web-assets/nikoncom/company/local/global/en/ir/ir_library/result/pdf/2026/26_all_e.pdf#page=8
13. A All about light and lasers in lithography. ASML · Accessed 9 September 2026
www.asml.com/en/technology/lithography-principles/light-and-lasers
14. A DUV Immersion Lithography: The Foundation of the US AI Advantage. Center for Technology & Statecraft · September 2026
techstatecraft.org/duv/
15. A The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology (CSET) · 21 January 2021 · p. 4
cset.georgetown.edu/wp-content/uploads/The-Semiconductor-Supply-Chain-Issue-Brief-1.pdf#page=4
16. A Export Controls on Semiconductor Manufacturing Items. Federal Register (Commerce Department; Industry and Security Bureau) · 25 October 2023
www.federalregister.gov/documents/2023/10/25/2023-23049/export-controls-on-semiconductor-manufacturing-items
17. A Expansion of End-User Controls To Cover Affiliates of Certain Listed Entities. Federal Register (Commerce Department; Industry and Security Bureau) · 30 September 2025
www.federalregister.gov/documents/2025/09/30/2025-19001/expansion-of-end-user-controls-to-cover-affiliates-of-certain-listed-entities
18. A 6 crucial steps in semiconductor manufacturing. ASML · 4 October 2023
www.asml.com/en/news/stories/2021/semiconductor-manufacturing-process-steps
19. A Photronics, annual report on Form 10-K for the fiscal year ended 31 October 2025. U.S. Securities and Exchange Commission (filing by Photronics) · 17 December 2025
www.sec.gov/Archives/edgar/data/810136/000114036125045801/ef20057458_10k.htm
20. A TWINSCAN NXE:3400B: EUV lithography systems. ASML · Accessed 27 September 2026
www.asml.com/en/products/euv-lithography-systems/twinscan-nxe3400b
21. A Stronger than expected demand drives ASML Q1 sales. U.S. Securities and Exchange Commission (filing by ASML Holding) · 19 April 2017
www.sec.gov/Archives/edgar/data/937966/000093796617000010/pressreleasequarterlyresul.htm
22. A ASML EUV lithography systems. ASML · Accessed 9 September 2026
www.asml.com/en/products/euv-lithography-systems
23. A See ASML's DUV lithography systems. ASML · Accessed 9 September 2026
www.asml.com/en/products/duv-lithography-systems
24. A EUV lithography: weighing the options. imec · 6 June 2019
www.imec-int.com/en/imec-magazine/imec-magazine-june-2019/euv-lithography-weighing-the-options-for-future-logic-and-memory-applications
25. A Applied Materials' Innovative Pattern-Shaping Technology Reduces the Cost, Complexity and Environmental Impact of Advanced Chip Manufacturing. Applied Materials · 28 February 2023
ir.appliedmaterials.com/news-releases/news-release-details/applied-materials-innovative-pattern-shaping-technology-reduces
26. A Strategic report: From ASML's Annual Report 2025. ASML · 24 February 2026 · p. 16
ourbrand.asml.com/m/8ab959d4926657b/original/asml-2025-annual-report-strategic-report-section.pdf#page=16
27. A Explore ASML San Diego. ASML · Accessed 9 September 2026
www.asml.com/en/company/about-asml/locations/san-diego
28. A Gigaphoton, company profile. Gigaphoton · Accessed 9 September 2026
www.gigaphoton.com/company
29. A Statement regarding export control regulations Dutch government. ASML · 30 June 2023
www.asml.com/en/news/press-releases/2023/statement-regarding-export-control-regulations-dutch-government
30. A Strategic report: From ASML's Annual Report 2025. ASML · 24 February 2026 · p. 30
ourbrand.asml.com/m/8ab959d4926657b/original/asml-2025-annual-report-strategic-report-section.pdf#page=30
31. A Strategic report: From ASML's Annual Report 2025. ASML · 24 February 2026 · p. 17
ourbrand.asml.com/m/8ab959d4926657b/original/asml-2025-annual-report-strategic-report-section.pdf#page=17
32. A 5 things you should know about High NA in EUV. ASML · 25 January 2024
www.asml.com/en/company/stories/2024/5-things-high-na-euv
33. A Financial performance: From ASML's Annual Report 2025. ASML · 24 February 2026 · p. 7
ourbrand.asml.com/m/419103cb23dfeaa4/original/asml-2025-annual-report-financial-performance-section.pdf#page=7
34. A Intel Foundry and ASML Accelerate Industry Readiness for High-NA EUV. Intel · 7 September 2026
www.intel.com/content/www/us/en/newsroom/news/intel-foundry/intel-foundry-asml-accelerate-industry-readiness-for-high-na-euv.html
35. A High NA EUV reaches new readiness milestone with first high-volume Logic product. ASML · 15 July 2026
www.asml.com/en/news/press-releases/2026/high-na-euv-reaches-new-readiness-milestone
36. A ASML and TSMC Announce Initiative to Pioneer Industry Transition to Large-Format Photomasks for High NA EUV. TSMC · 8 September 2026
pr.tsmc.com/english/news/3338
37. A ASML reports €11.7 billion total net sales and €5.0 billion net income in Q3 2026. U.S. Securities and Exchange Commission (filing by ASML Holding) · 14 October 2026
www.sec.gov/Archives/edgar/data/937966/000162828026065813/pressreleasefinancialresul.htm

38. A EUV lithography and technology. Carl Zeiss · Accessed 9 September 2026
www.zeiss.com/semiconductor-manufacturing-technology/inspiring-technology/euv-lithography.html
39. A Good things come in ever-smaller packages. TRUMPF · Accessed 9 September 2026
www.trumpf.com/en_US/products/lasers/euv-drive-laser/
40. A VDL ETG builds complex frames for Zeiss. VDL Groep · 22 March 2019
web.archive.org/web/20240726075938/https://www.vdlgroep.com/en/news/vdl-etg-builds-complex-frames-for-zeiss
41. A Sourcing & Supply Chain. ASML · Accessed 27 October 2026
www.asml.com/en/careers/teams/sourcing-and-supply-chain
42. A Financial Results: The Year Ended March 31, 2026. Nikon · 8 May 2026 · p. 6
www.nikon.com/content/dam/web-assets/nikoncom/company/local/global/en/ir/ir_library/result/pdf/2026/26_all_e.pdf#page=6
43. A Semiconductor Lithography Equipment. Canon · Accessed 18 October 2026
global.canon/en/product/indtech/semicon/
44. A Canon delivers FPA -1200NZ2C nanoimprint lithography system for semiconductor manufacturing to the Texas Institute for Electronics. Canon · 26 September 2024
global.canon/en/news/2024/20240926.html
45. A The Netherlands expands export control measure for advanced semiconductor manufacturing equipment. Government of the Netherlands · 6 September 2024
www.government.nl/latest/news/2024/09/06/the-netherlands-expands-export-control-measure-advanced-semiconductor-manufacturing-equipment
46. A Statement regarding partial revocation export license. ASML · 1 January 2024
www.asml.com/en/news/press-releases/2023/statement-regarding-partial-revocation-export-license
47. A Klever: export controls on advanced semiconductor manufacturing equipment to be tightened. Government of the Netherlands · 15 January 2025
www.government.nl/latest/news/2025/01/15/klever-export-controls-on-advanced-semiconductor-manufacturing-equipment-to-be-tightened
48. A One Year Suspension of Expansion of End-User Controls for Affiliates of Certain Listed Entities. Federal Register (Commerce Department; Industry and Security Bureau) · 12 November 2025
www.federalregister.gov/documents/2025/11/12/2025-19846/one-year-suspension-of-expansion-of-end-user-controls-for-affiliates-of-certain-listed-entities
49. A Strategic report: From ASML's Annual Report 2025. ASML · 24 February 2026 · p. 40
ourbrand.asml.com/m/8ab959d4926657b/original/asml-2025-annual-report-strategic-report-section.pdf#page=40
50. A Strategic report: From ASML's Annual Report 2025. ASML · 24 February 2026 · p. 65
ourbrand.asml.com/m/8ab959d4926657b/original/asml-2025-annual-report-strategic-report-section.pdf#page=65
51. A Strategic report: From ASML's Annual Report 2025. ASML · 24 February 2026 · p. 68
ourbrand.asml.com/m/8ab959d4926657b/original/asml-2025-annual-report-strategic-report-section.pdf#page=68

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